

Testing magnitude-weighted spherical k-means

The whole pipeline — GIM features, PCA embed, k-means, softpart bank, ablation — run 3 times on Llama-3.2-1B at C=256 with an 8k-position pilot, fitting 4 variants per run on identical features. The decisive control is running the CURRENT method twice with different k-means init: that alone swings the Gini from 0.56 to 0.89 and the largest single-component effect from 0.12 to 2.56 nats, a 22× range. Against that, magnitude weighting moves nothing (Welch t = +0.13, +0.10, −0.18 on the three metrics). Total study: 4 minutes per run.

